Topic 15: Random Variables

02-680: Essentials of Mathematics and Statistics

October 29, 2026

The formal definition of a random variable X, is as a mapping:

$$X : \Omega \to \mathbb{R}$$

which assigns a real number $X(\omega)$ to an outcome in our space (note this is not a probability in and of itself).

As an example, let's say we toss a coin twice, we know we get $\Omega_{twocoin}$ as defined in previous lectures. We can assign a variable $C(\omega)$ to be the count of the number of heads in each outcome.

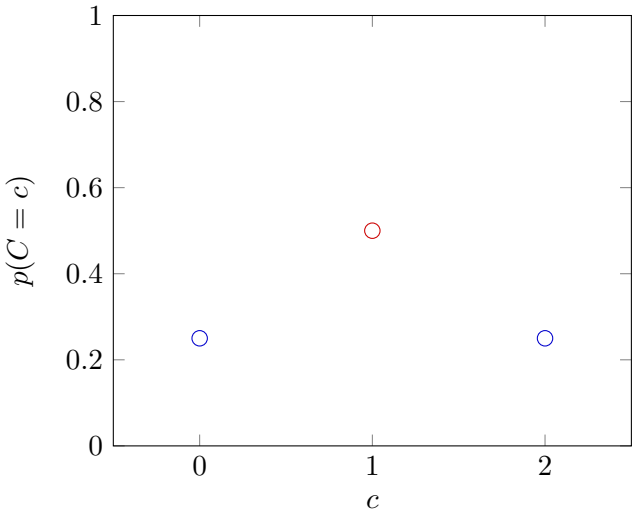
We can then examine probabilities of the possible values a random variable X can take on (its range), we usually use a lower case of the same name, x. We can think of this as the sum of the probabilities of all the outcomes that lead to a particular value.

ω	$p(\omega)$	$C(\omega)$
HH	1/4	2
HT	1/4	1
TH	1/4	1
TT	1/4	0

c	$p(C = c)$
0	1/4
1	1/2
2	1/4

1 Discrete Random Variables — Probability Mass Functions

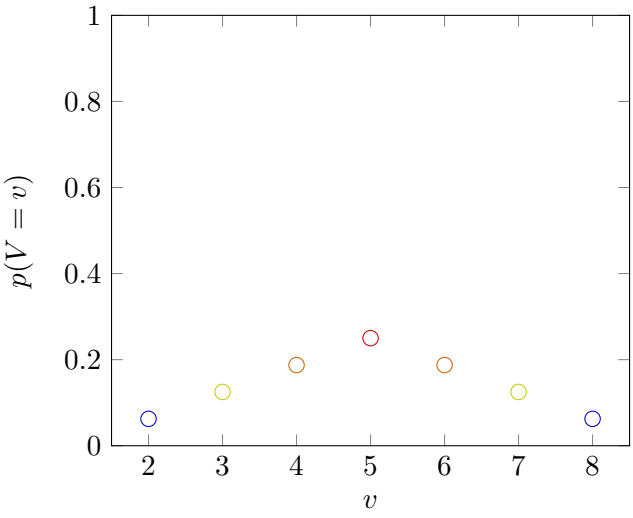
In this case we call the probability distribution a ***probability mass function*** (PMF). Below is a visualization of the PMF for C above.



As another example let's again examine rolling two dice (for simplicity, 4 sided), and let $V(\omega)$ be the value of the sum of the two rolled dice.

ω	$p(\omega)$	$V(\omega)$
11	1/16	2
12	1/16	3
13	1/16	4
14	1/16	5
21	1/16	3
22	1/16	4
23	1/16	5
24	1/16	6
31	1/16	4
32	1/16	5
33	1/16	6
34	1/16	7
41	1/16	5
42	1/16	6
43	1/16	7
44	1/16	8

v	$p(V = v)$
2	1/16
3	1/8
4	3/16
5	1/4
6	3/16
7	1/8
8	1/16



1.1 Bernoulli

A well-known PMF is a ***Bernoulli Distribution*** which is defined as follows with probability of success ($k = 1$) $0 \leq \alpha \leq 1$ for $\Omega = \{0, 1\}$:

$$p(k) = \begin{cases} \alpha & k = 1 \\ 1 - \alpha & k = 0 \end{cases}$$

or put another way

$$p(k) = \alpha^k (1 - \alpha)^{(1-k)} \quad \forall k \in \Omega$$

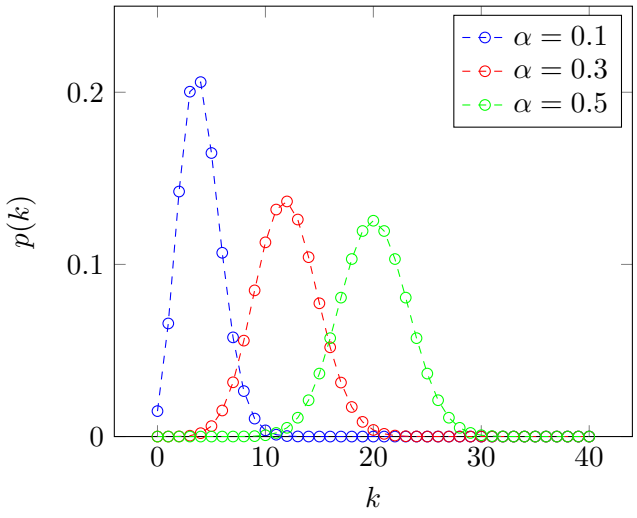
Example Scenario. A single (possibly biased) coin flip with probability of heads α.

1.2 Binomial

Another (maybe the most) well-known PMF is the ***Binomial Distribution***, again we say the probability of a (single) trial's success is $0 \leq \alpha \leq 1$, but now we have n trials and we want to know the probability of getting $k \leq n$ successes (this makes $\Omega = \{0, 1, 2, ..., n\}$). In this case

$$p(k) = \binom{n}{k} \alpha^k (1 - \alpha)^{n-k} \quad \forall k \in \Omega.$$

Example Scenario. n repeated coin flips, each one being independent and having probability of heads α.



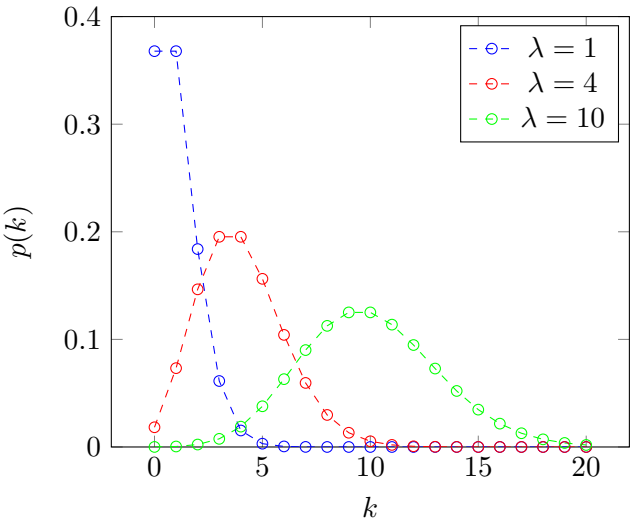
1.3 Poisson

Poisson distributions are used to model rare events over time, with λ representing the expected number of events per interval.

$$p(k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \forall k \in \Omega \text{ where } \Omega = \{0, 1, 2, ...\}.$$

Example Scenario. One of the first applications of the Poisson distribution was by statistician Ladislaus Bortkiewicz. In the late 1800s, he investigated accidental deaths by horse kick of soldiers in the Prussian army. He analyzed 20 years of data for 10 army corps, equivalent to 200 years of observations of one corps.

He found that a mean of 0.61 soldiers per corps died from horse kicks each year. However, most years, no soldiers died from horse kicks. On the other end of the spectrum, one tragic year there were four soldiers in the same corps who died from horse kicks.*

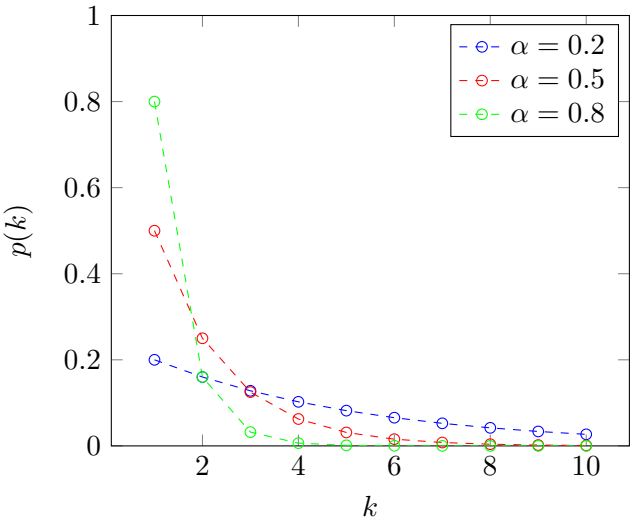


1.4 Geometric

$$p(k) = (1 - \alpha)^{k-1}\alpha \quad \forall k \in \Omega \text{ where } \Omega = \{1, 2, ...\}.$$

Example Scenario. Probability that the k-th coin-flip is the first head (probability of heads α) with the previous throws being tails. ~~Number of coin flips (probability of heads α) needed to get k heads.~~

*see **Ladislaus von Bortkiewicz** "The Law of Small Numbers", 1898. pp. 23-25



2 Continuous Random Variables — Probability Density Functions

In the case of continuous random variable (think grades, weight, height, rainfall [in.], etc.) the probability distribution is referred to as a ***Probability Density Function*** (PDF). We still have the same properties as before, everything integrates (sums) to 1, all the densities are non-negative, etc.; but we typically consider a *range* rather than a single value to test. In the case of PDFs the probability is the area under the curve along a range (the total area is going to end up being one), but when we're talking about area under the curve we're really considering the integral:

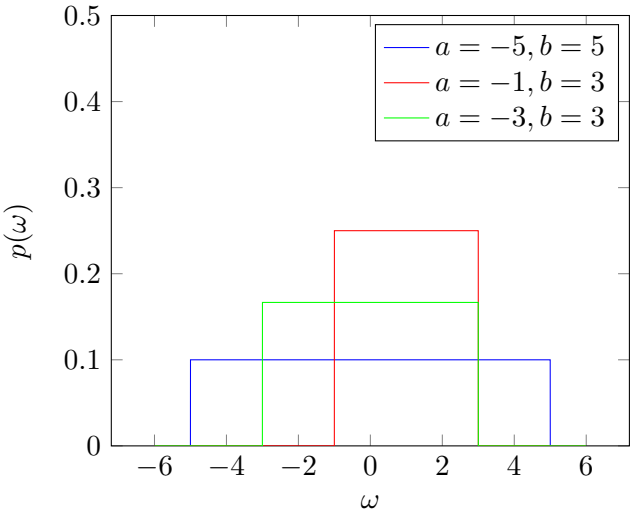
$$p(a \leq X \leq b) = \int_a^b f(x)dx$$

Notice that $f(x)$ is not a probability in and of itself, the probability comes from the integral. That's why the probability of a single value ($p(X = x)$) is (usually) zero.

2.1 Uniform

All outcomes with value between a and $b > a$ are equally likely:

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

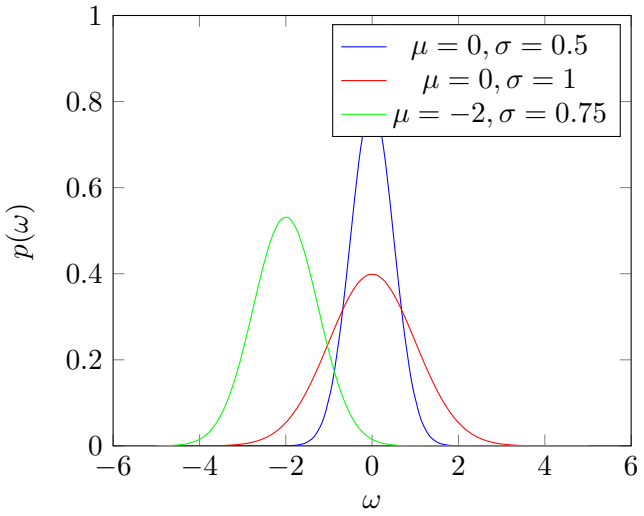


2.2 Gaussian

Also called the ***normal*** distribution. It is parameterized by the mean (center) value of the samples $\mu \in \mathbb{R}$, and the standard deviation (spread) of the samples $\sigma > 0$.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \; e^{\frac{(x-\mu)^2}{-2\sigma^2}}$$

Example Scenario. Most natural phenomena (average daily temperature, height of adults, etc.).

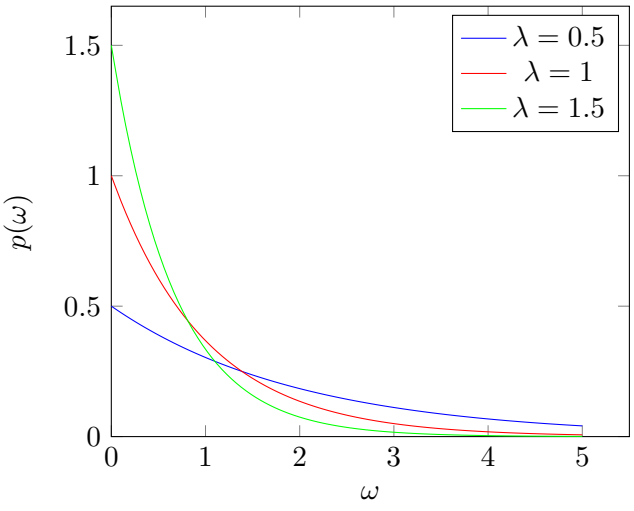


2.3 Exponential

Parameterized by a value of $\lambda > 0$, which is the average rate of arrivals over time (assuming $x \geq 0$).

$$f(x) = \lambda e^{-x\lambda}$$

Example Scenario. The ***waiting time*** between rare events (notice this can also be thought of as the time between Poisson events, Sec. 1.3).



3 Cumulative Distribution Functions

Sometimes we need to know the probability over a range, where the lower bound is $-\infty$ (or upper/∞). Something like:

$$p(X \leq x).$$

3.1 Discrete

For discrete this is as follows:

$$p(X \leq x) = \sum_{y \in \text{range}(X), y \leq x} p(X = y)$$

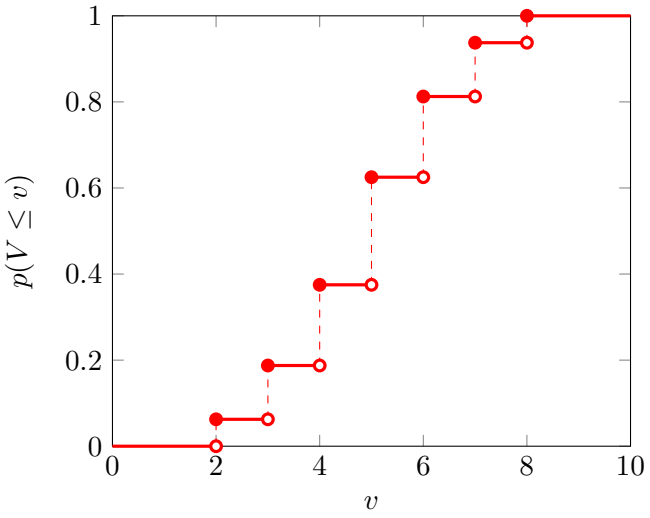
In the example above, the probability

$$p(V \leq v) = \sum_{d \in \{x \in \mathbb{Z} | x \geq 2 \wedge x \leq v\}} p(V = d)$$

so if we want to know

$$p(V \leq 4) = \sum_{d \in \{2,3,4\}} p(V = d) = p(V = 2) + p(V = 3) + p(V = 4) = \frac{1}{16} + \frac{1}{8} + \frac{3}{16} = \frac{3}{8}$$

This in and of itself can also be visualized:



Notice that technically, the CDF even for a discrete value is defined between elements in the range, so in the example above we can ask $p(V \leq 4.78) = \frac{3}{8}$.

3.2 Continuous

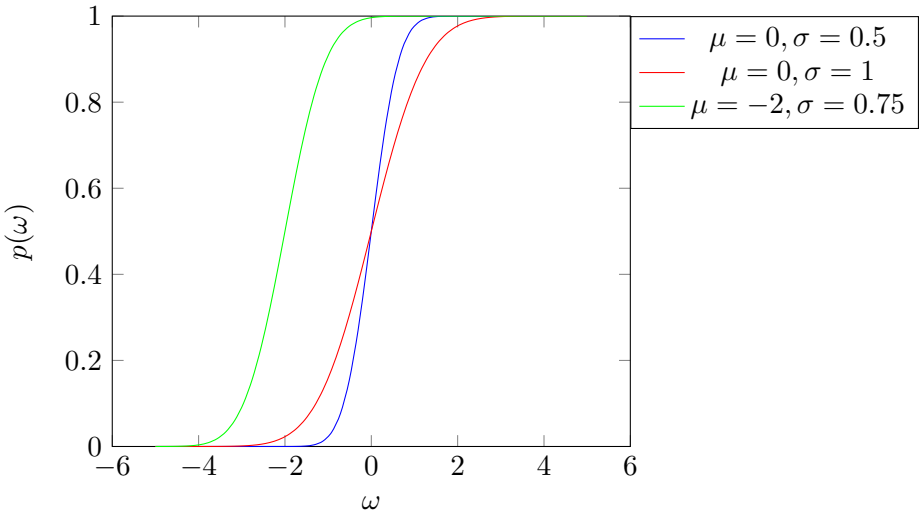
For continuous random variables, it looks similar to the property we had earlier:

$$p(X \leq x) = \int_{-\infty}^{x} f(z)dz$$

So for instance, if we look at the Gaussian distribution:

$$p(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} \; e^{\frac{(z-\mu)^2}{-2\sigma^2}} dz$$

Note that the actual integration here is beyond the scope of this course.



Useful References

Wasserman. "All of Statistics: A Concise Course in Statistical Inference" §§2.1–2.4